InnoLink: A Knowledge Graph and LLM-based Recommendation System for University–Expert–Enterprise Collaboration

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Abstract

Bridging the gap between academic institutions, domain experts, and enterprises is essential for fostering innovation and technology transfer, particularly in developing countries such as Vietnam. However, existing approaches to matchmaking among these stakeholders are predominantly manual and lack scalability. In this paper, we present **InnoLink**, an integrated recommendation system that combines (1) a semi-automated web crawling and entity extraction pipeline with human-in-the-loop validation, (2) a domain-specific knowledge graph capturing the tripartite relationships among universities/research institutes, experts, and enterprises, (3) a graph-based smart matching algorithm leveraging node embeddings for computing compatibility scores, and (4) a large language model (LLM) module that generates natural language explanations for each recommendation. We evaluate InnoLink on a dataset of **[DATA: N]** profiles collected from **[DATA: describe regions/sources]** and demonstrate that the knowledge graph-based matching outperforms keyword-based and TF-IDF baselines in terms of Precision@5 and NDCG@10. Human evaluation of LLM-generated explanations confirms their factual accuracy and usefulness. Our system offers a practical, explainable, and scalable solution for strengthening university–industry collaboration in Vietnam.

Keywords: knowledge graph, recommendation system, university–industry collaboration, large language model, explainable recommendation, web crawling

1 Introduction

The collaboration between universities, research institutes, and enterprises plays a critical role in driving innovation, facilitating technology transfer, and promoting socio-economic development. In Vietnam, the government has recognized the strategic importance of strengthening university–industry linkages, as reflected in Resolution No. 57 of the Politburo on science, technology, and innovation **[CITE: Resolution 57]**. Despite these policy efforts, the rate of research commercialization remains low. A survey of 570 faculty members at five Vietnamese technical universities found that university–industry linkages are relatively weak due to differences in research objectives, geographical constraints, and limited information sharing [1]. Similarly, a recent comprehensive review confirmed that collaboration modes in Vietnam remain concentrated in informal, low-commitment channels such as sponsorships and internships, with limited adoption of systematic matching mechanisms [2].

A fundamental challenge is that the process of connecting the right expert or institution with the right enterprise demand is still predominantly manual—relying on personal networks, ad hoc conferences, and fragmented databases. There is no unified platform that can automatically discover, profile, and match these three types of stakeholders based on their actual competencies and needs.

To address this gap, we propose **InnoLink**, an integrated recommendation system built upon four key components:

1. A **semi-automated data collection pipeline** that crawls institutional and enterprise websites, extracts structured profiles using LLM-based entity extraction, and incorporates human-in-the-loop validation to ensure data quality.
2. A **domain-specific knowledge graph** (KG) with a tailored ontology that captures three entity types—universities/research institutes, experts, and enterprises—along with their attributes and inter-relationships.
3. A **smart matching engine** that computes compatibility scores between entities using graph-based similarity measures and node embedding techniques (Node2Vec).
4. An **LLM-based explanation generator** that produces natural language justifications for each recommendation, enhancing transparency and user trust.

The main contributions of this paper are as follows:

- We design and implement a semi-automated pipeline for collecting and validating stakeholder profiles from heterogeneous web sources, achieving an extraction F1-score of **[DATA: X%]** before human review and **[DATA: Y%]** after review.
- We propose a tripartite knowledge graph ontology specifically designed for the Vietnamese academic–industry ecosystem, together with a hybrid matching algorithm that combines structural graph similarity and semantic embeddings.
- We integrate an LLM module that generates explainable recommendations in Vietnamese, and evaluate its output quality through human assessment, demonstrating that it enhances user trust and decision-making.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 presents the system architecture. Sections 4, 5, and 6 detail the three core modules. Section 7 reports experimental results. Section 8 concludes the paper.

2 Related Work

2.1 Expert and Collaboration Recommendation Systems

Expert recommendation systems aim to identify individuals with relevant expertise for a given task or query. A comprehensive survey by Husain et al. [4] found that approximately 65% of expert finding systems focus on the academic domain, employing methods ranging from information retrieval to graph-based approaches. In the scholarly context, recommendation systems have been developed for finding collaborators, reviewers, and research papers [5]. For instance, Gollapalli et al. proposed a content-based system that computes similarity between researchers based on profiles extracted from publications and academic homepages **[CITE: Gollapalli et al.]**. The REMatch system [6] leverages NLP, machine learning, and interactive visualization to match experts to research proposals.

More recently, the topic-institution graph approach proposed by [7] demonstrated the use of SciBERT-based embeddings to capture latent semantic correlations in university–industry collaboration networks. However, most existing systems focus on bilateral matching (e.g., researcher–paper, expert–query) and do not address the tripartite matching problem among institutions, experts, and enterprises that is central to our work.

2.2 Knowledge Graph-based Recommendation

Knowledge graphs have emerged as a powerful substrate for recommendation systems due to their ability to encode rich semantic relationships between entities [8]. KG-based approaches generally fall into three categories: embedding-based methods that learn low-dimensional representations of entities and relations, path-based methods that mine multi-hop connectivity patterns, and GNN-based methods that aggregate neighborhood information through message passing [9]. The integration of KGs with collaborative filtering, as in collaborative knowledge graphs (CKGs), has shown promise in alleviating data sparsity and cold-start problems [10].

Despite these advances, domain-specific ontology design for academic–industry ecosystems—particularly in the Vietnamese context—remains unexplored. Existing KG-based recommenders primarily target e-commerce, entertainment, or general academic citation networks, lacking the tripartite entity structure required for our scenario.

2.3 LLMs for Explainable Recommendation

The rise of large language models has opened new avenues for generating natural language explanations in recommendation systems. A recent systematic review by Said [11] covering publications from November 2022 to November 2024 identified only six studies directly addressing LLM-based explanation of recommendations, highlighting the nascency of this research direction. Among these, Silva et al. showed that ChatGPT-generated explanations improved user engagement through contextually relevant justifications **[CITE: Silva et al. 2024 IUI]** The XRec framework [12] proposed a collaborative relation tokenizer that transforms user-item relationships into latent embeddings, enabling LLMs to generate explanations grounded in collaborative filtering signals.

Our work differs from prior studies in several ways: (1) we target a domain-specific tripartite matching scenario rather than general item recommendation; (2) we use open-source LLMs (e.g., Qwen, Llama) running locally via Ollama rather than proprietary APIs; and (3) we evaluate explanation quality specifically for Vietnamese language output.

3 System Architecture

[FIGURE: Insert Figure 1: Overall system architecture diagram showing the four-layer p

InnoLink consists of four interconnected layers, as illustrated in Figure **??**.

- **Data Collection Layer**: Web crawlers (built with Scrapy/Selenium) extract raw content from institutional websites, Google Scholar profiles, and enterprise portals. An LLM-based extraction module (using Qwen3/Llama3.2 via Ollama) converts unstructured text into structured JSON profiles. A human-in-the-loop validation interface (Streamlit/Gradio) allows domain experts to review and correct extracted information.
- **Knowledge Graph Layer**: Validated profiles are ingested into a knowledge graph stored in **[TODO: Neo4j or NetworkX—confirm which]**. The KG is constructed according to a domain-specific ontology (detailed in Section 5).
- **Smart Matching Engine**: Given a query entity (e.g., an enterprise seeking expertise in IoT), the engine computes compatibility scores against candidate entities using a combination of graph-structural similarity and Node2Vec embeddings.
- **LLM Explanation Layer**: For the top-k recommended matches, the LLM generates a natural language explanation describing why the match is relevant, based on the profiles and matching scores of both parties.

The technology stack is summarized in Table 1.

Table 1: Technology stack of InnoLink.

Component	**Technology**
Web Crawling	Scrapy, Selenium
Entity Extraction	Qwen3 / Llama3.2 via Ollama
Knowledge Graph	**[TODO: Neo4j / NetworkX]**
Node Embedding	Node2Vec (gensim / PyTorch Geometric)
Matching Algorithm	Cosine similarity + weighted scoring
LLM Explanation	Qwen3 / Llama3.2 via Ollama
User Interface	Streamlit / Gradio

4 Semi-Automated Data Collection

4.1 Data Sources and Crawling Strategy

We target three categories of web sources for profile construction:

1. **University/institute websites**: Faculty pages, research group pages, and publication lists from Vietnamese universities. **[TODO: List specific universities crawled.]**
2. **Academic databases**: Google Scholar profiles and DBLP pages of researchers, providing publication records, citation counts, and co-author networks.
3. **Enterprise portals**: Company websites, business registration databases, and industry association pages, providing information on business sectors, technology needs, and partnership history.

Each source type requires a tailored crawling strategy due to structural heterogeneity. We use Scrapy for structured/semi-structured pages and Selenium for JavaScript-rendered content. **[TODO: Add details on rate limiting, deduplication, and data volume.]**

4.2 LLM-based Entity Extraction

Rather than relying on rule-based NER or pre-trained NER models—which have limited coverage for Vietnamese domain-specific terminology—we employ an LLM-based approach. Given a raw text snippet from a crawled web page, we prompt the LLM to extract structured information in JSON format. The prompt template is shown below:

```
Given the following text from a [university/enterprise] website, extract
the following fields as JSON:
- name:  string
- type:  "university" | "expert" | "enterprise"
- research_fields:  list of strings
- skills_technologies:  list of strings
- publications:  list of {title, year, venue}
- location:  string
- contact:  string
If a field is not found, return null.
Text:  [RAW_TEXT]
```

[TODO: Add a concrete example showing raw input text and extracted JSON output

4.3 Human-in-the-Loop Validation

Extracted profiles are presented to domain experts through a validation interface built with **[TODO: Streamlit/Gradio]** . For each profile, the reviewer can:

- Accept the extraction without changes;
- Make minor corrections (e.g., fixing field names, adding missing skills);
- Reject and re-extract with a modified prompt.

We measure the effectiveness of the LLM extraction by computing precision, recall, and F1-score for each field type, both before and after human review (Table 2).

Table 2: Entity extraction quality per field type (before and after human review).

Field	Before Review P	Before Review R	Before Review F1	After Review P	After Review R	After Review F1
Name	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:
Research fields	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:
Skills/Technologies	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:
Publications	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:
Location	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:
Average	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:]	[DATA:

[DATA: Report: acceptance rate without corrections, minor correction rate, major co

5 Knowledge Graph and Smart Matching

5.1 Ontology Design

We design a domain-specific ontology for the Vietnamese academic–industry ecosystem. The ontology defines three primary entity types and their relationships, as illustrated in Figure ??.

[FIGURE: Insert Figure 2: Ontology diagram showing entity types (University/Instit

5.1.1 Entity Types and Attributes

- **University/Institute** ($\mathcal{U}$): name, type (public/private), location, research focus areas, number of researchers, notable labs/centers.
- **Expert** ($\mathcal{E}$): name, affiliation, research fields, skills/technologies, publications, h-index, years of experience.
- **Enterprise** ($\mathcal{D}$): name, industry sector, location, technology needs, partnership history, company size.

5.1.2 Relationship Types

- $\mathcal{E} \xrightarrow{\texttt{affiliated\_with}} \mathcal{U}$: An expert belongs to a university/institute.
- $\mathcal{E} \xrightarrow{\texttt{researches}} \mathcal{F}$: An expert works in a research field $\mathcal{F}$.
- $\mathcal{D} \xrightarrow{\texttt{needs\_technology}} \mathcal{F}$: An enterprise requires expertise in field $\mathcal{F}$.
- $\mathcal{U} \xrightarrow{\texttt{has\_expert}} \mathcal{E}$: An institution hosts an expert.
- $\mathcal{D} \xrightarrow{\texttt{collaborates\_with}} \mathcal{U}$: An enterprise has a collaboration history with an institution.
- $\mathcal{E} \xrightarrow{\texttt{has\_skill}} \mathcal{S}$: An expert possesses a specific skill $\mathcal{S}$.

[TODO: Consider adding: temporal attributes (collaboration year), strength weights

5.2 Knowledge Graph Construction

Given the validated profiles from Section 4, we construct the knowledge graph using the following pipeline:

1. **Entity node creation**: Each profile is mapped to a node with attributes populated from the extracted JSON.
2. **Field/skill node creation**: Research fields and skills are normalized using a controlled vocabulary **[TODO: describe the vocabulary source—Vietnamese NAFOSTED** and added as intermediate nodes.
3. **Edge creation**: Relationships are established based on extracted data (e.g., affiliation links from expert profiles) and inferred connections (e.g., two experts share a research field node).

[DATA: Report KG statistics: number of nodes per type, number of edges per type, g

5.3 Smart Matching Algorithm

Given a query entity q (e.g., an enterprise) and a set of candidate entities $\mathcal{C}$ (e.g., experts), the matching algorithm computes a compatibility score $\text{Score}(q, c)$ for each candidate $c \in \mathcal{C}$.

5.3.1 Graph-structural Similarity

We compute similarity based on shared neighbors in the knowledge graph. Let $\mathcal{N}(v)$ denote the set of field/skill nodes connected to entity v. The Jaccard-based field similarity is:

$$\text{sim}_{\text{field}}(q, c) = \frac{|\mathcal{N}(q) \cap \mathcal{N}(c)|}{|\mathcal{N}(q) \cup \mathcal{N}(c)|} \tag{1}$$

5.3.2 Embedding-based Similarity

We apply Node2Vec [13] to learn d-dimensional embeddings for all nodes in the KG. Node2Vec performs biased random walks on the graph parameterized by return parameter p and in-out parameter q, then trains a Skip-Gram model to produce embeddings. The embedding-based similarity is computed as cosine similarity:

$$\text{sim}_{\text{emb}}(q, c) = \frac{\mathbf{z}_q \cdot \mathbf{z}_c}{\|\mathbf{z}_q\| \cdot \|\mathbf{z}_c\|} \tag{2}$$

where $\mathbf{z}_v \in \mathbb{R}^d$ is the Node2Vec embedding of node v.

5.3.3 Geographic Proximity

For practical relevance, we incorporate a geographic proximity factor:

$$\text{sim}_{\text{geo}}(q, c) = \exp\left(-\frac{d_{\text{geo}}(q, c)}{\sigma}\right) \tag{3}$$

where $d_{\text{geo}}(q, c)$ is the geographic distance between q and c, and σ is a scaling parameter. **[TODO: Determine σ empirically or set as province-level granularity.]**

5.3.4 Combined Score

The final matching score is a weighted combination:

$$\text{Score}(q, c) = \alpha \cdot \text{sim}_{\text{field}}(q, c) + \beta \cdot \text{sim}_{\text{emb}}(q, c) + \gamma \cdot \text{sim}_{\text{geo}}(q, c) \tag{4}$$

where $\alpha + \beta + \gamma = 1$. **[TODO: Describe how weights are tuned—grid search, expert judgm**
The top-k candidates with the highest scores are returned as recommendations.

6 LLM-based Recommendation Explanation

6.1 Prompt Design

For each recommended pair (q, c) with matching score and detailed sub-scores, we construct a prompt that provides the LLM with sufficient context to generate a meaningful explanation:

```
You are an expert advisor on university-industry collaboration. Given the
following profiles and matching analysis, write a concise explanation (3-5
sentences) in Vietnamese describing why this match is beneficial for both
parties.

Enterprise profile: [JSON_ENTERPRISE]
Expert profile: [JSON_EXPERT]
Matching scores:
- Field overlap: [sim_field]
- Embedding similarity: [sim_emb]
- Geographic proximity: [sim_geo]
- Overall score: [Score]

Explain why this expert is a good match for this enterprise's needs. Be
specific about overlapping competencies and potential collaboration opportunities.
```

6.2 LLM Selection and Comparison

We compare three open-source LLMs running locally via Ollama for Vietnamese language generation quality:

- **Qwen3** **[TODO: specify parameter size]**
- **Llama3.2** **[TODO: specify parameter size]**
- **Gemma** **[TODO: specify parameter size]**

Each model generates explanations for the same set of **[DATA: N]** recommended pairs. Quality is assessed through human evaluation (Section 6.3).

6.3 Human Evaluation Protocol

Three evaluators **[TODO: describe evaluators—domain experts, faculty members, industry** independently rate each LLM-generated explanation on three criteria using a 5-point Likert scale:

1. **Factual accuracy**: Does the explanation correctly reflect the information in the profiles?
2. **Usefulness**: Does the explanation help a decision-maker understand why this match is valuable?
3. **Language naturalness**: Is the Vietnamese text fluent and professional?

Inter-rater agreement is measured using Fleiss' Kappa.

7 Experiments

7.1 Dataset

[DATA: Describe the dataset:Number of university/institute profiles Number of expert p

Table 3: Human evaluation of LLM-generated explanations (mean Likert scores, 1–5).

Model	**Accuracy**	**Usefulness**	**Naturalness**
Qwen3	**[DATA:]**	**[DATA:]**	**[DATA:]**
Llama3.2	**[DATA:]**	**[DATA:]**	**[DATA:]**
Gemma	**[DATA:]**	**[DATA:]**	**[DATA:]**
Fleiss' κ	**[DATA:]**	**[DATA:]**	**[DATA:]**

Table 4: Dataset and knowledge graph statistics.

Metric	**Value**
University/institute profiles	**[DATA:]**
Expert profiles	**[DATA:]**
Enterprise profiles	**[DATA:]**
Research field nodes	**[DATA:]**
Skill/technology nodes	**[DATA:]**
Total KG nodes	**[DATA:]**
Total KG edges	**[DATA:]**
Average node degree	**[DATA:]**

7.2 Matching Performance

We compare our KG-based matching approach against two baselines:

- **Keyword Matching**: Direct string matching between enterprise technology needs and expert research fields/skills.
- **TF-IDF + Cosine**: TF-IDF vectorization of textual profile descriptions followed by cosine similarity ranking.

Evaluation is conducted via expert assessment: **[DATA: N]** evaluators rate the relevance of the top-k recommendations for **[DATA: M]** query entities. We report Precision@5, Precision@10, and NDCG@10.

Table 5: Matching performance comparison.

Method	**P@5**	**P@10**	**NDCG@10**
Keyword Matching	**[DATA:]**	**[DATA:]**	**[DATA:]**
TF-IDF + Cosine	**[DATA:]**	**[DATA:]**	**[DATA:]**
KG Jaccard only	**[DATA:]**	**[DATA:]**	**[DATA:]**
KG + Node2Vec	**[DATA:]**	**[DATA:]**	**[DATA:]**
KG + Node2Vec + Geo	**[DATA:]**	**[DATA:]**	**[DATA:]**

7.3 Ablation Study

To understand the contribution of each component in the combined score (Eq. 4), we conduct an ablation study by varying the weights α, β, and γ.

[DATA: Report ablation results: effect of removing geographic proximity, effect of dif

7.4 Case Study

We illustrate the end-to-end system with two concrete scenarios:

7.4.1 Scenario 1: Enterprise Seeking IoT Expertise

[DATA: Describe: An enterprise in [industry] needs IoT solutions. The system returns t

[FIGURE: Insert Figure 3: Screenshot of the InnoLink UI showing a recommendation

7.4.2 Scenario 2: Research Institute Seeking Industry Partner

[DATA: Describe: A research institute specializing in [field] seeks enterprise partners for

7.5 Discussion

Advantages: Compared to manual matching, InnoLink provides scalable, reproducible, and explainable recommendations. The KG structure enables multi-hop reasoning (e.g., finding an expert through their institution's existing industry partnerships). The LLM explanation module enhances transparency and user trust.

Limitations:

- **Data dependency**: The system's quality depends heavily on the completeness and freshness of crawled web data. Many Vietnamese institutions have incomplete or outdated websites.
- **Cold start**: New entities with few connections in the KG receive less accurate recommendations.
- **LLM cost**: Running LLMs locally requires GPU resources. Response latency may be an issue for real-time applications.
- **Scale**: The current evaluation is limited to **[DATA: N]** profiles. Larger-scale validation is needed.

Future directions:

- Incorporating user feedback loops to refine matching scores over time (collaborative filtering).
- Integrating RAG (Retrieval-Augmented Generation) to allow the LLM to query the KG directly for richer explanations.
- Expanding data sources to include patent databases, grant records, and social media profiles.

8 Conclusion

This paper presented InnoLink, an integrated recommendation system for connecting universities, experts, and enterprises. The system combines semi-automated web crawling with human validation, a tripartite knowledge graph with domain-specific ontology, graph-based smart matching using Node2Vec embeddings, and LLM-powered explainable recommendations. Experimental results on a dataset of **[DATA: N]** profiles demonstrate that the KG-based approach achieves a **[DATA: X%]** improvement in Precision@5 over TF-IDF baselines, and that LLM-generated explanations receive an average usefulness score of **[DATA: Y]** /5 from human evaluators. InnoLink offers a practical solution for strengthening university–industry collaboration, particularly in the Vietnamese context where such connections remain underdeveloped. Future work will focus on scaling the system, integrating user feedback, and exploring RAG-based explanation generation.

Acknowledgments

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